Team Presentation Playbook: Tasks & Speaking Points Matrix

This playbook outlines a detailed work division for your 5-member team. It maps Concrete Codebase Tasks and Speaking Points across 4 Development Phases, giving each member a clear script of what they "planned, built, and verified" during the project lifecycle.

The 5-Department Work Division

Department 1: Core Backend & Algorithm Engineering

- Focus: Relational database schemas, API architecture, data models, and fuzzy-matching logic.
- Key Files:
- matching_strategy.py (RapidFuzz metrics calculations)
- identity.py (SQLAlchemy Database models)
- resolution_service.py (Resolution coordinator)

Department 2: UI/UX & Frontend Development

- Focus: Responsive user interfaces, interactive charts, page routing, and dashboard visualizations.
- Key Files:
- Dashboard.jsx (Chart.js stats widgets)
- IdentityRegistration.jsx (Multi-step ingestion wizard)
- index.css (CSS Design Tokens & themes)

Department 3: Cybersecurity & Identity Access Management (IAM)

- Focus: JWT security, Token rotations, password hashes, middleware decorators, and tenant boundaries.
- Key Files:
- AuthContext.jsx (Frontend token validation & session handlers)
- auth_service.py (Authentication database services)
- middleware.py (Flask rate limiter & organization permission filters)

Department 4: Product Operations & Auditing Systems

- Focus: Manual resolution checks, verification workflow desk, audit log exports, and settings controllers.
- Key Files:
- VerificationDetails.jsx (Operator action forms)
- AuditLogs.jsx (Event timelines & diff visualizer)
- SystemSettings.jsx (Dynamic threshold configs)

Department 5: DevOps & Site Reliability Engineering (SRE)

- Focus: Multi-stage container builds, local reverse proxies, databases configuration, and automation scripts.
- Key Files:
- Dockerfile (Multi-stage backend production environment builder)
- nginx.conf (HSTS, CSP, and rate limiting proxy setup)
- backup_restore.sh (Shell cron backing tool)

Phase-by-Phase Tasks & Speaking Script

Phase 1: Requirements Analysis & Schema Design

Department 1 (Backend & Algorithms)

- What was Built: Designed the database schema mapping relational tables for identities, clusters, organizations, and user logs.
- Speaking Points: *"I designed the database schemas, mapped the primary foreign keys for users, organizations, and identities, and configured the baseline SQLAlchemy models to support multi-source digital credentials."*

Department 2 (UI/UX & Frontend)

- What was Built: Configured the global CSS style design system tokens and set up the React Router tree inside `App.jsx`.
- Speaking Points: *"I designed the global CSS system, configured style variables for dark/light themes, and mapped out the page layout wireframes using React Router configurations."*

Department 3 (Cybersecurity & IAM)

- What was Built: Created the baseline permissions array list (`user:write`, `identity:ingest`, `identity:resolve`, `system:settings`) within database seeds.
- Speaking Points: *"I defined the Role-Based Access Control (RBAC) permission structure and planned the token-based security flow, specifying permissions like `identity:ingest` and `identity:resolve`."*

Department 4 (Operations & Auditing)

- What was Built: Defined the state lifecycle transition workflow for incoming verification records (PENDING, COMPLETED, FAILED).
- Speaking Points: *"I defined the state machine for manual reviews, specifying the attributes needed to compare duplicates (e.g. name, phone number, and birthdates)."*

Department 5 (DevOps & SRE)

- What was Built: Initialized the MySQL/PostgreSQL services inside `docker-compose.yml` and configured local environment variables.
- Speaking Points: *"I set up the local developer environment, configured a Git repository with feature branches for each member, and scaffolded the initial database compose services."*

Phase 2: Core Module Development

Department 1 (Backend & Algorithms)

- What was Built: Coded Jaro-Winkler and Levenshtein token sorting similarity algorithms within `matching_strategy.py`.
- Speaking Points: *"I built the backend matching engines, integrating the RapidFuzz token sorting metrics (Jaro-Winkler and Levenshtein) to link records across Aadhaar and ABHA."*

Department 2 (UI/UX & Frontend)

- What was Built: Created the core analytics Dashboard (integrating Chart.js metrics) and developed the multi-step identity ingestion forms.
- Speaking Points: *"I built the main Dashboard console with Chart.js trackers and constructed the step-by-step identity ingestion wizard, supporting temporary drafts."*

Department 3 (Cybersecurity & IAM)

- What was Built: Implemented user login bcrypt hashing, organization data isolation checks, and JWT access/refresh token rotation.
- Speaking Points: *"I implemented JWT access/refresh token rotation handlers, hashed user passwords with bcrypt, and built Flask decorators to block cross-organization data leaks."*

Department 4 (Operations & Auditing)

- What was Built: Developed the Verification requests listing workspace and built the System Settings console to modify threshold scores dynamically.
- Speaking Points: *"I built the Verification Desk dashboard listing pending reviews, and created a dynamic system settings UI to tweak threshold variables in real-time."*

Department 5 (DevOps & SRE)

- What was Built: Built multi-stage frontend/backend Dockerfiles utilizing non-root users (`USER appuser`) for vulnerability hardening.
- Speaking Points: *"I wrote multi-stage, non-root user Dockerfiles for the frontend and backend applications to minimize our security footprint."*

Phase 3: Hardening & Auditing Systems

Department 1 (Backend & Algorithms)

- What was Built: Optimized database lookup speeds by implementing indexes on deterministic sha256 identifier columns.
- Speaking Points: *"I optimized the resolution query times by indexing the deterministic cryptographic hashes of identity attributes, bringing execution speeds under 100ms."*

Department 2 (UI/UX & Frontend)

- What was Built: Polish responsive dashboard styles and mock-tested Canvas rendering containers inside tests.
- Speaking Points: *"I designed the custom data visualization charts mapping resolution confidence levels and integrated responsive mobile views across all consoles."*

Department 3 (Cybersecurity & IAM)

- What was Built: Coded backend JWT Blocklist models to revoke tokens on logout and built frontend `PermissionGuard` wrappers.
- Speaking Points: *"I configured the JWT blocklist database model to invalidate tokens on logout and built frontend PermissionGuard hooks to protect private routes."*

Department 4 (Operations & Auditing)

- What was Built: Developed Audit log event details drawers, JSON/CSV data exporters, and side-by-side data diff highlighting viewers.
- Speaking Points: *"I built the Audit Logs viewer, adding detailed diff comparison highlights, user activity tables, and dynamic CSV/JSON export actions."*

Department 5 (DevOps & SRE)

- What was Built: Configured Nginx proxy routing variables, implementing rate limit metrics and custom CSP headers.
- Speaking Points: *"I configured the Nginx reverse proxy with Content-Security-Policy (CSP) headers, HSTS parameters, and DDoS rate-limiting rules."*

Phase 4: Integration, Testing & Deployment

Department 1 (Backend & Algorithms)

- What was Built: Wrote integration test cases in backend/tests validating correct identity clusters split/merge cascades.
- Speaking Points: *"I wrote unit tests for the resolution engines, verifying that identity merges trigger appropriate updates across clusters."*

Department 2 (UI/UX & Frontend)

- What was Built: Configured React Query states queries hooks and wrapped routing components in test context wrappers.
- Speaking Points: *"I integrated React Query to cache API requests, and mock-tested all charts using virtualized canvas contexts."*

Department 3 (Cybersecurity & IAM)

- What was Built: Verified the user account lockout policy checks and added frontend session expiry modal warnings countdowns.
- Speaking Points: *"I verified the login lockout behavior, ensuring that accounts are locked for 15 minutes after 5 consecutive failed login attempts."*

Department 4 (Operations & Auditing)

- What was Built: Checked and tested dynamic CSV/JSON generation, ensuring user profiles database records export perfectly.
- Speaking Points: *"I tested the CSV export APIs, verifying that user tables dump clean formats containing all data mapping elements."*

Department 5 (DevOps & SRE)

- What was Built: Wrote the automated DB backup shell script `backup_restore.sh` and provisioned initial database seeds.
- Speaking Points: *"I wrote an automated database backup utility script, created the database seed files, and deployed the final integrated staging containers."*